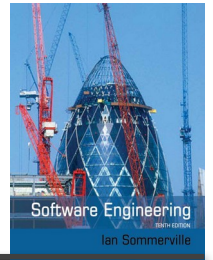


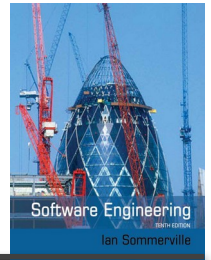
Chapter 8 – Software Testing

Topics covered



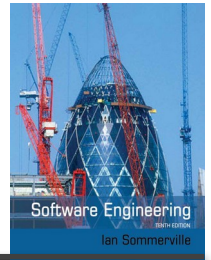
- ✧ Development testing
- ✧ Release testing
- ✧ User testing

Program testing



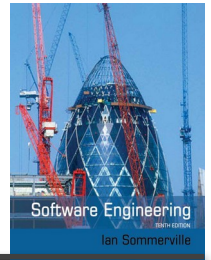
- ✧ Testing is intended to show that **a program does what it is intended to do and to discover program defects** before it is put into use.
- ✧ When you test software, you execute a program using **artificial data**.
- ✧ You check the results of the test run for **errors, anomalies or information about the program's non-functional attributes**.
- ✧ Can reveal the **presence of errors NOT their absence**.
- ✧ Testing is part of a more general verification and validation process, which also includes static validation techniques.

Program testing goals



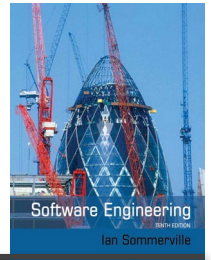
- ✧ To demonstrate to the developer and the customer that the software **meets its requirements**.
 - For **custom software**, this means that there should be at least one test for **every requirement** in the requirements document. For **generic software** products, it means that there should be tests for **all of the system features**, plus **combinations** of these features, that will be incorporated in the product release.
- ✧ To discover situations in which the behavior of the software is **incorrect, undesirable or does not conform to its specification**.
 - Defect testing is concerned with rooting out undesirable system behavior such as system crashes, unwanted interactions with other systems, incorrect computations and data corruption.

Validation and defect testing



- ✧ The first goal leads to validation testing
 - You expect the system to perform correctly using a given set of test cases that reflect the system's expected use.
- ✧ The second goal leads to defect testing
 - The test cases are designed to expose defects. The test cases in defect testing can be deliberately obscure and need not reflect how the system is normally used.

Testing process goals



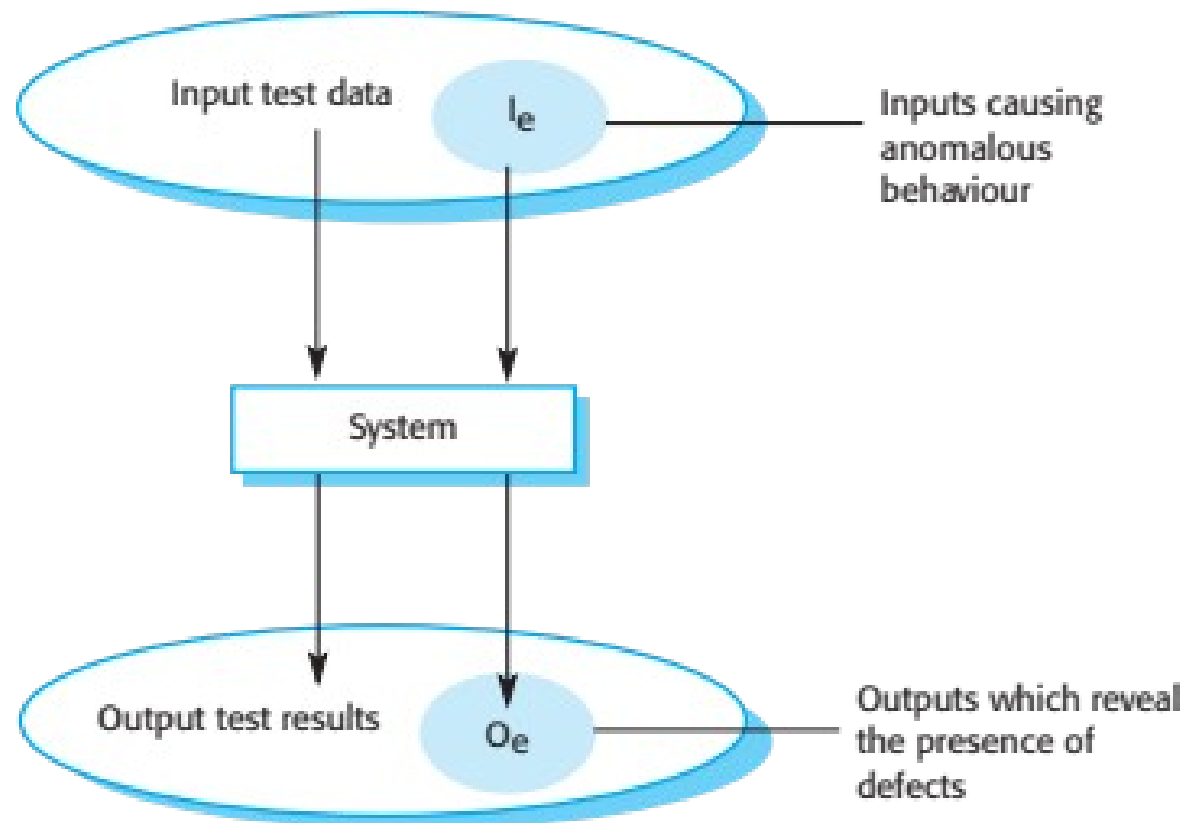
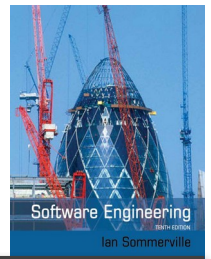
✧ Validation testing

- To demonstrate to the developer and the system customer that the software meets its requirements
- A successful test shows that the system operates as intended.

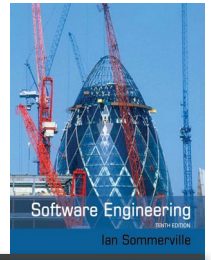
✧ Defect testing

- To discover faults or defects in the software where its behaviour is incorrect or not in conformance with its specification
- A successful test is a test that makes the system perform incorrectly and so exposes a defect in the system.

An input-output model of program testing

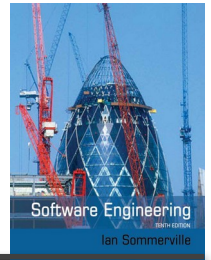


Verification vs validation



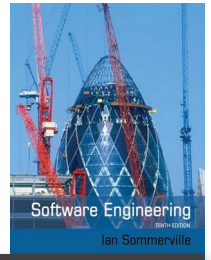
- ✧ Verification:
 - "Are we building the product right".
- ✧ The software should conform to its specification.
- ✧ Validation:
 - "Are we building the right product".
- ✧ The software should do what the user really requires.

V & V confidence



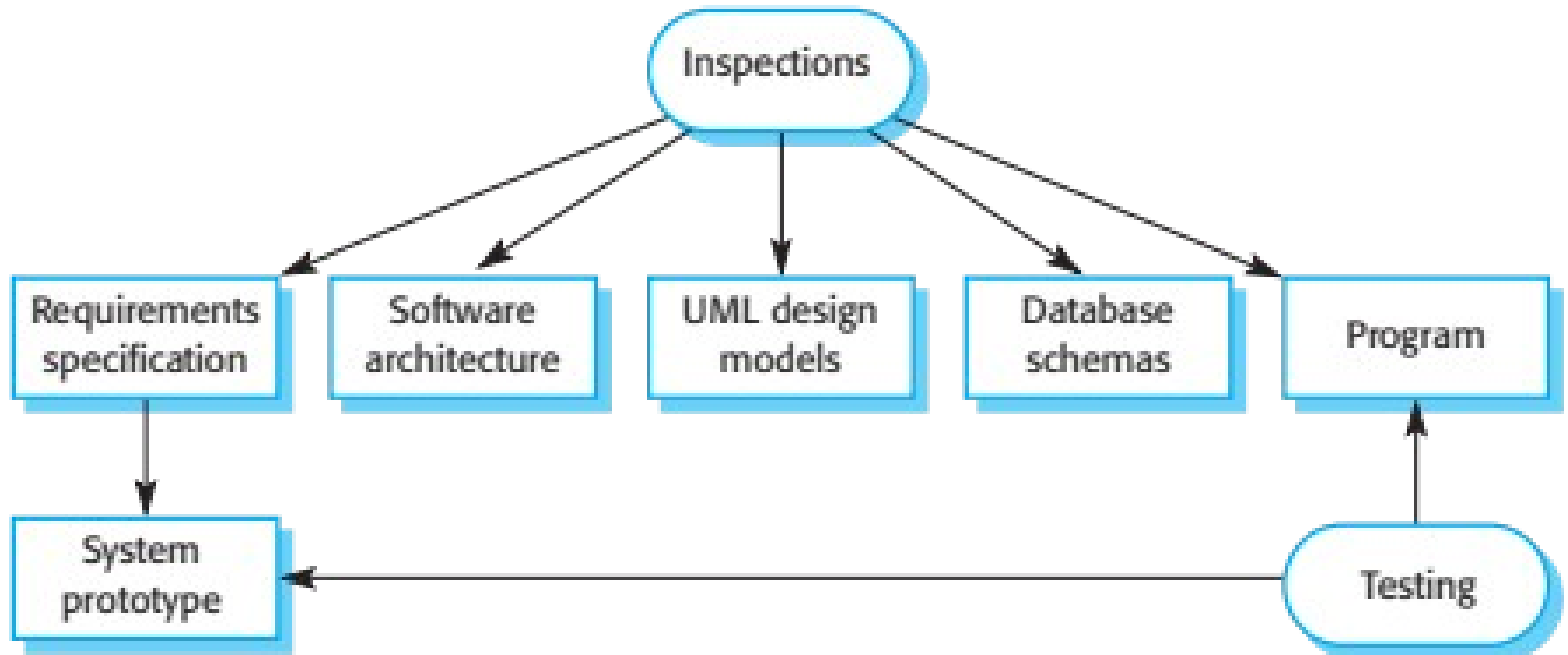
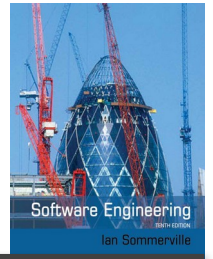
- ✧ Aim of V & V is to establish confidence that the system is 'fit for purpose'.
- ✧ Depends on system's purpose, user expectations and marketing environment
 - Software purpose
 - The level of confidence depends on how critical the software is to an organisation.
 - User expectations
 - Users may have low expectations of certain kinds of software.
 - Marketing environment
 - Getting a product to market early may be more important than finding defects in the program.

Inspections and testing

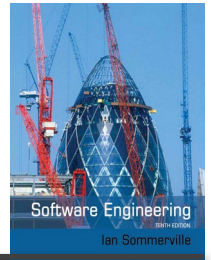


- ✧ **Software inspections** Concerned with analysis of the static system representation to discover problems (static verification)
 - May be supplement by tool-based document and code analysis.
- ✧ **Software testing** Concerned with exercising and observing product behaviour (dynamic verification)
 - The system is executed with test data and its operational behaviour is observed.

Inspections and testing

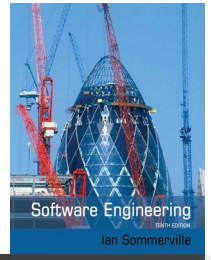


Software inspections



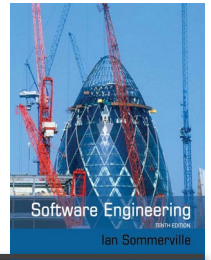
- ✧ These involve people examining the source representation with the aim of discovering anomalies and defects.
- ✧ Inspections **not require execution** of a system so may be used before implementation.
- ✧ They may be applied to **any representation** of the system (requirements, design, configuration data, test data, etc.).
- ✧ They have been shown to be an **effective technique** for discovering program errors.

Advantages of inspections



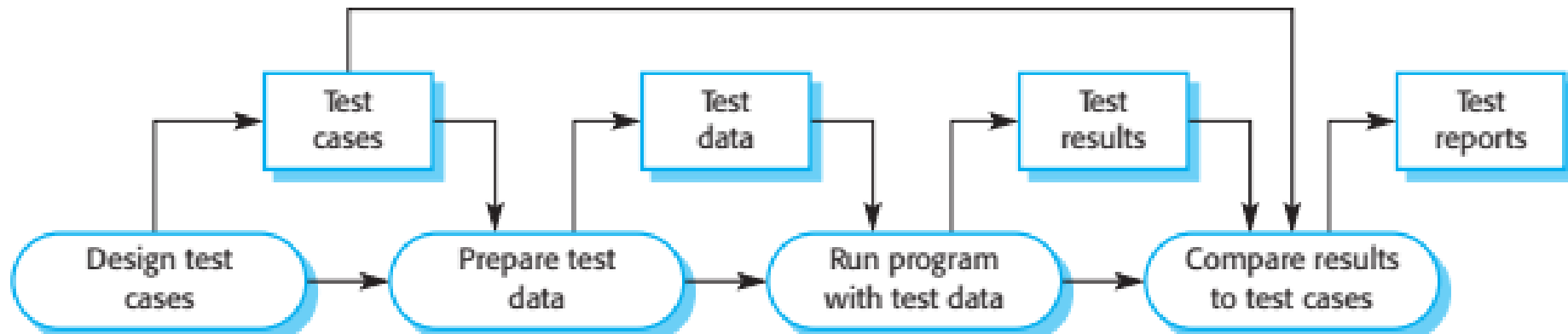
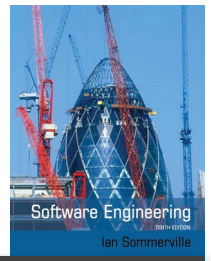
- ✧ During testing, **errors can mask (hide) other errors**.
Because inspection is a static process, you don't have to be concerned with interactions between errors.
- ✧ **Incomplete versions of a system** can be inspected without additional costs. If a program is incomplete, then you need to develop specialized test harnesses to test the parts that are available.
- ✧ As well as searching for program defects, an inspection can also consider **broader quality attributes** of a program, such as compliance with standards, portability and maintainability.

Inspections and testing

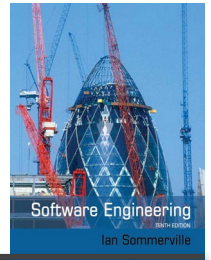


- ✧ Inspections and testing are **complementary** and not opposing verification techniques.
- ✧ Both should be used during the V & V process.
- ✧ Inspections can check conformance with a specification but **not conformance with the customer's real requirements**.
- ✧ Inspections **cannot check some non-functional** characteristics such as performance, usability, etc.

A model of the software testing process



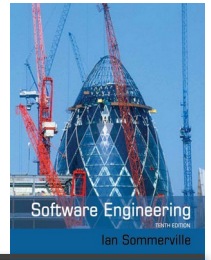
Stages of testing



- ✧ Development testing, where the system is tested **during development** to discover bugs and defects.
- ✧ Release testing, where a separate testing team test a complete version of the system **before it is released** to users.
- ✧ User testing, where users or potential users of a system test the system **in their own environment**.

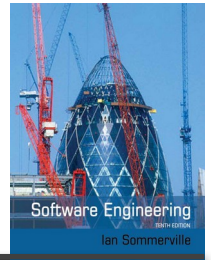
Development testing

Development testing



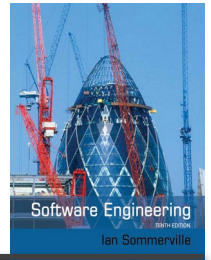
- ✧ Development testing includes all testing activities that are carried out by the team developing the system.
 - Unit testing, where individual program units or object classes are tested. Unit testing should focus on testing the functionality of **objects or methods**.
 - Component testing, where several individual units are integrated to create **composite components**. Component testing should focus on testing component **interfaces**.
 - System testing, where some or all of the components in a system are integrated and the system is **tested as a whole**. System testing should focus on testing component interactions.

Unit testing



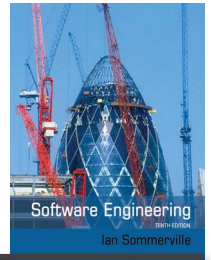
- ✧ Unit testing is the process of testing individual components in isolation.
- ✧ It is a defect testing process.
- ✧ Units may be:
 - Individual functions or methods within an object
 - Object classes with several attributes and methods
 - Composite components with **defined interfaces** used to access their functionality.

Automated testing



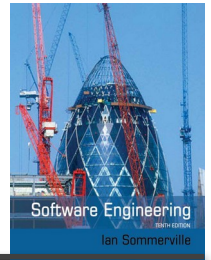
- ✧ Whenever possible, unit testing should be **automated** so that tests are run and checked without manual intervention.
- ✧ In automated unit testing, you make use of a test **automation framework** (such as JUnit) to write and run your program tests.

General testing guidelines



- ✧ Choose inputs that force the system to **generate all error messages**
- ✧ Design inputs that cause **input buffers to overflow**
- ✧ **Repeat the same input** or series of inputs numerous times
- ✧ Force **invalid outputs** to be generated
- ✧ Force computation results to be **too large** or **too small**.

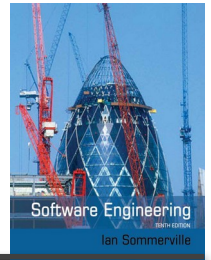
Testing policies



- ✧ **Exhaustive** system testing is **impossible** so testing policies which define the required system test **coverage** may be developed.
- ✧ Examples of testing policies:
 - All system functions that are accessed through menus should be tested.
 - Combinations of functions (e.g. text formatting) that are accessed through the same menu must be tested.
 - Where user input is provided, all functions must be tested with both correct and incorrect input.

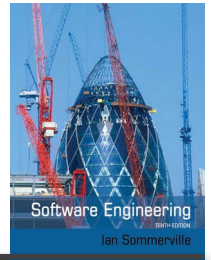
Release testing

Release testing

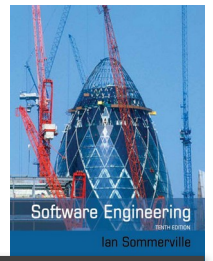


- ✧ Release testing is the process of testing a **particular release** of a system that is intended for use outside of the development team.
- ✧ The primary goal of the release testing process is to convince the supplier of the system that it is **good enough for use**.
 - Release testing, therefore, has to show that the system delivers its specified functionality, performance and dependability, and that it does not fail during normal use.
- ✧ Release testing is usually a **black-box testing process** where tests are only derived from the system specification.

Performance testing

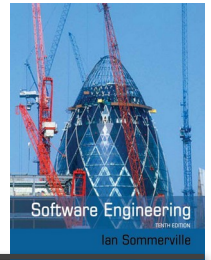


- ✧ Part of release testing may involve testing the emergent properties of a system, such as performance and reliability.
- ✧ Tests should reflect the **profile of use of the system**.
- ✧ Performance tests usually involve planning a series of tests where the **load is steadily increased** until the system performance becomes unacceptable.
- ✧ **Stress testing** is a form of performance testing where the system is deliberately overloaded to test its failure behaviour.



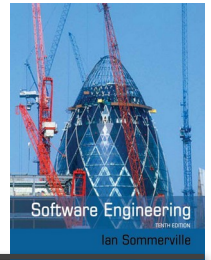
User testing

User testing



- ✧ User or customer testing is a stage in the testing process in which **users or customers provide input** and advice on system testing.
- ✧ User testing is essential, even when comprehensive system and release testing have been carried out.
 - The reason for this is that influences from the user's working environment have a major effect on the reliability, performance, usability and robustness of a system. These cannot be replicated in a testing environment.

Types of user testing



✧ Alpha testing

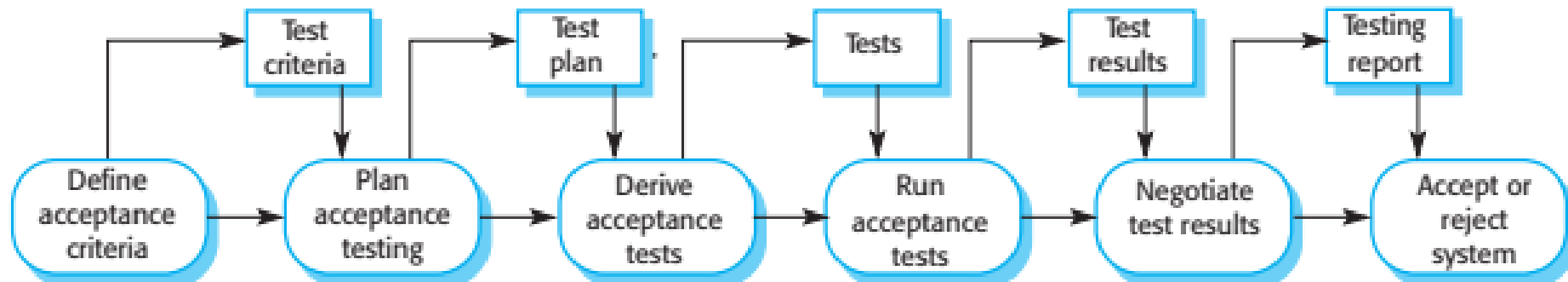
- Users of the software work with the development team to test the software at the developer's site.

✧ Beta testing

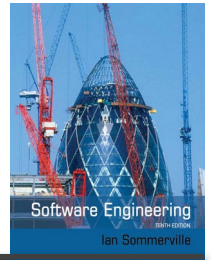
- A release of the software is made available to users to allow them to experiment and to raise problems that they discover with the system developers.

✧ Acceptance testing

- Customers test a system to decide whether or not it is ready to be accepted from the system developers and deployed in the customer environment. Primarily for custom systems.

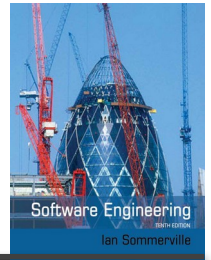


Key points



- ✧ Testing can only show the presence of errors in a program. It cannot demonstrate that there are no remaining faults.
- ✧ Development testing is the responsibility of the software development team. A separate team should be responsible for testing a system before it is released to customers.
- ✧ Development testing includes unit testing, in which you test individual objects and methods component testing in which you test related groups of objects and system testing, in which you test partial or complete systems.

Key points



- ✧ When testing software, you should try to 'break' the software by using experience and guidelines to choose types of test case that have been effective in discovering defects in other systems.
- ✧ Wherever possible, you should write automated tests. The tests are embedded in a program that can be run every time a change is made to a system.
- ✧ Test-first development is an approach to development where tests are written before the code to be tested.
- ✧ Scenario testing involves inventing a typical usage scenario and using this to derive test cases.
- ✧ Acceptance testing is a user testing process where the aim is to decide if the software is good enough to be deployed and used in its operational environment.